DESIGN
PRESENTATION

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PROBLEM DEFINITION

Urban authorities lack a reliable method to identify physically vulnerable urban areas from satellite imagery. Dense, poorly structured zones can worsen hazard impacts, including rapid disease spread. This project proposes an AI-based system to classify urban areas into low, medium, and high vulnerability using observable physical and environmental features.

PURPOSE AND NEED

- Urban vulnerability arises from factors such as high building density, informal settlements, and environmental risks.
- Traditional assessment methods rely on census data, surveys, and field inspections, which can be costly or outdated.
- These methods often lack fine spatial resolution and timely availability.
- Satellite imagery provides a scalable, consistent, and globally accessible source of urban data.
- Using AI-based analysis on satellite imagery enables a systematic and repeatable identification of vulnerable urban areas.

PROJECT OBJECTIVES

- The primary objective of the project is to spatially identify and classify urban vulnerability.
- Classification is based on absolute, predefined physical risk conditions.
- Each spatial unit is evaluated independently without reference to surrounding areas.
- Risk levels reflect directly observed vulnerability rather than relative comparison across regions.

LITERATURE SURVEY

[1] X. Jean-Francois, A. Taubenböck, T. Esch, et al., "Urban Land Cover Classification Using Deep Neural Networks Based on VHR Multi-Spectral Image and Point Cloud Integration," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2024.

Relevance:

This paper shows that using deep learning with high-resolution satellite images and 3D point cloud data gives better land cover results than traditional methods. It explains how combining different data sources helps clearly identify buildings, roads, and vegetation in cities

LITERATURE SURVEY

[2] H. Taubenböck,
“Remote Sensing for the Analysis of Global Urbanization,” 2019.

Relevance:
This work explains how satellite data, night-time light images, and socio-economic information can be combined to study how cities grow and change at different scales. It shows that using multiple data sources helps better understand urban growth patterns and internal city structures.

LITERATURE SURVEY

[3] H. Taubenböck, et al.,
“Slums from Space—15 Years of Slum Mapping Using Remote Sensing,” Remote Sensing, 2016.

Relevance:

This review summarizes 15 years of research on mapping slums using high-resolution satellite images. It shows that machine learning, texture analysis, and OBIA work best, and explains why better and more general methods are needed for mapping slums at a global scale.

LITERATURE SURVEY

[4] S. Chakraborty, et al.,
“Mapping Urban Heat Vulnerability Using Remote Sensing and GIS,” 2019.

Relevance:

This study uses GIS and satellite data along with population information to map areas that are most vulnerable to urban heat. It shows that combining heat intensity with population characteristics helps identify high-risk heat hotspots in cities.

LITERATURE SURVEY

[5] B. A. Thiam, et al.,
“Deep Learning-based Land Use and Land Cover Changes Detection from Satellite Imagery: A Case Study of the City of Richard Toll,” 2024.

Relevance:

This study uses deep learning models like U-Net on satellite images from different years to detect land use and land cover changes. It shows that U-Net gives better and more detailed change maps, helping in automatic monitoring of urban and agricultural areas.

LITERATURE SURVEY

Paper Title	Relevance	Pros	Cons
Urban Land Cover Classification Using Deep Neural Networks Based on VHR Multi-Spectral Image and Point Cloud Integration (2024)	Shows that deep learning combined with high-resolution satellite images and point cloud data improves urban land cover classification.	• High accuracy compared to traditional OBIA• Uses both spectral and 3D information• Good performance in complex urban areas	• Requires point cloud data, which is not always available• Higher computational cost
Remote Sensing for the Analysis of Global Urbanization (2019)	Explains how multiple remote sensing and socio-economic datasets can be used to study global and local urban growth patterns.	• Covers urbanization at multiple scales• Integrates social and physical data• Strong conceptual frameworks	• No single algorithm or accuracy metrics• More descriptive than implementation-focused
Slums from Space—15 Years of Slum Mapping Using Remote Sensing (2016)	Reviews 15 years of methods for mapping slums using satellite imagery and highlights effective techniques.	• Comprehensive review of methods• Identifies best-performing techniques• Useful for understanding challenges	• No new model proposed• Limited focus on real-time or large-scale automation
Mapping Urban Heat Vulnerability Using Remote Sensing and GIS (2019)	Demonstrates how GIS and satellite data combined with population data can identify urban heat risk zones.	• Combines physical and social factors• Useful for urban planning• Spatially detailed vulnerability maps	• No machine learning comparison• Results depend heavily on indicator selection
Deep Learning-based Land Use and Land Cover Changes Detection from Satellite Imagery (2024)	Uses deep learning segmentation (U-Net) on multi-temporal images to detect land cover changes automatically.	• Automated change detection• Detailed class-wise results• U-Net performs better than FCN-8	• Accuracy metrics not clearly reported• Limited to a single study area

PROPOSED METHOD

1. Satellite Image Acquisition and Study Area Definition

A high-resolution satellite image covering the selected urban area is acquired from reliable sources and used as the sole input for vulnerability assessment.

2. Spatial Segmentation into Analysis Units

The satellite image is divided into uniform, non-overlapping spatial patches, each treated as an independent unit of analysis.

3. Feature Extraction per Risk Dimension

For each patch, physical indicators related to urban structure, dense or informal settlement patterns, and environmental exposure are extracted, with explicit identification of the contributing risk factors.

PROPOSED METHOD

4. Rule-Based Risk Evaluation

Extracted indicators are evaluated against predefined absolute vulnerability conditions using transparent, rule-based logic applied independently to each patch.

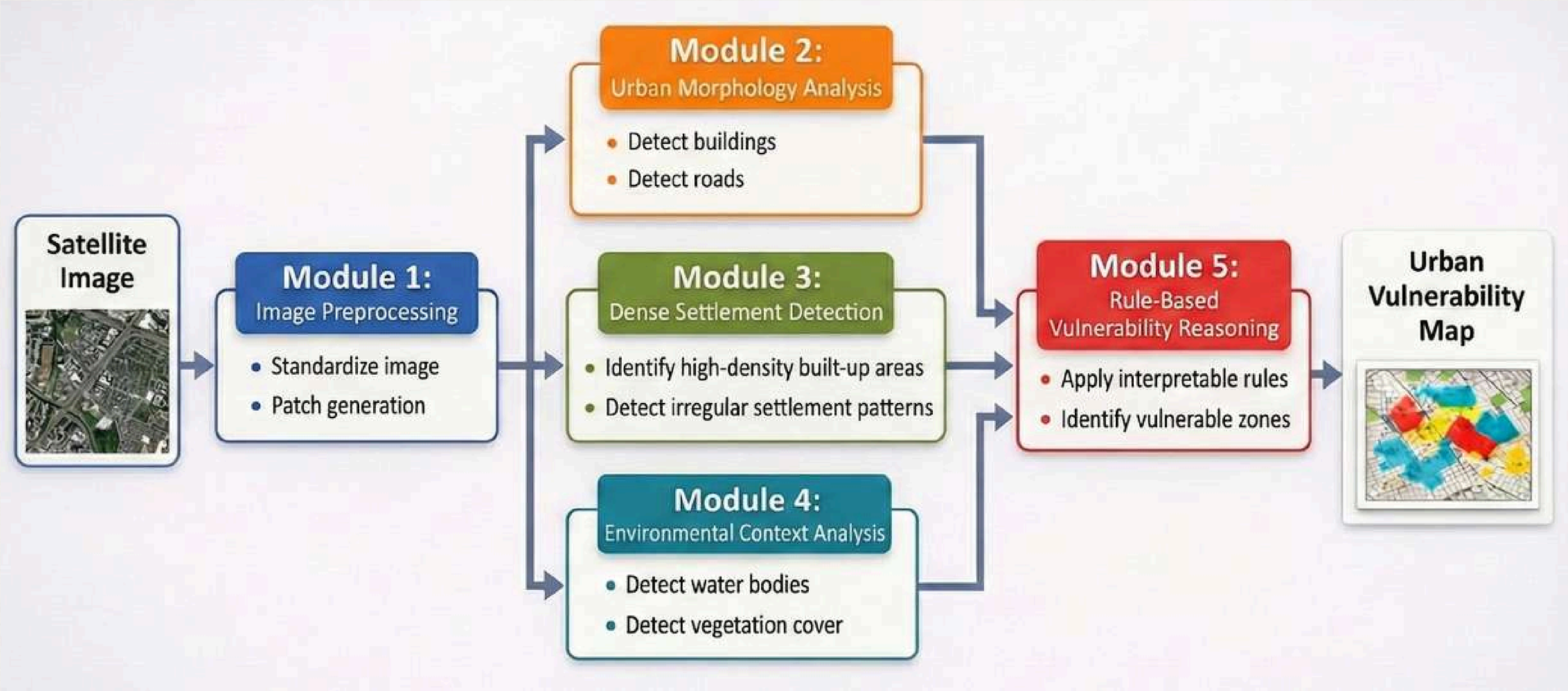
5. Absolute Risk Classification

Each patch is classified as low, medium, or high risk based on the number of vulnerability conditions detected, without relative comparison across regions.

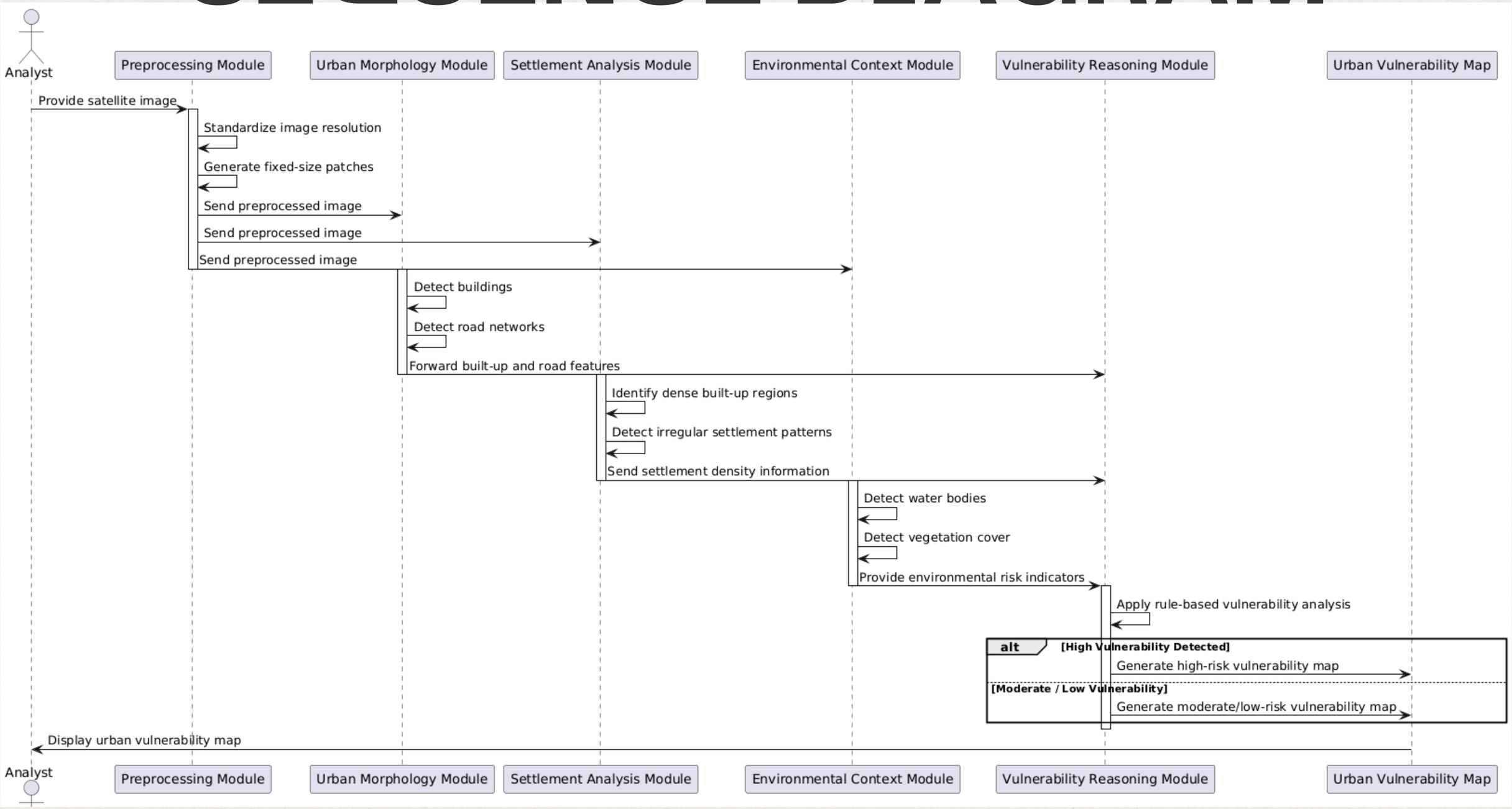
6. Spatial Risk Mapping and Interpretation

Risk dimension maps are combined to generate a final urban vulnerability

ARCHITECTURE DIAGRAM



SEQUENCE DIAGRAM



MODULES

Module 1: Image Preprocessing

Goal: Prepare satellite imagery for analysis

- Standardize image format: Resize, normalize, and align satellite images to ensure uniform input
- Patch generation: Divide large satellite images into smaller patches (e.g., 32×32 px) to simplify processing
- This enables efficient processing by deep learning models and downstream tasks

MODULES

Module 2: Urban Morphology Analysis

Goal: Identify built environment structure

- Detect buildings: Use edge detection, texture features, or pretrained models to identify building footprints
- Detect roads: Apply line-detection and segmentation to extract road networks
- These elements help understand urban form and accessibility

MODULES

Module 3: Dense Settlement Detection

Goal: Detect informal/slum-like regions

- High-density built-up areas: Identify clusters of small, tightly packed structures
- Irregular patterns: Detect chaotic or unplanned layouts that signal informal development
- Often done using deep learning (CNNs), texture analysis, or unsupervised clustering

MODULES

Module 4: Environmental Context Analysis

Goal: Assess environmental risks

- Water bodies: Detect proximity to rivers/lakes to assess flood risk
- Vegetation cover: Identify lack of green space, which can signal heat vulnerability
- Uses NDVI, color indices, or spectral bands in satellite data

MODULES

Module 5: Rule-Based Vulnerability Reasoning

Goal: Combine all data to create vulnerability zones

- Apply interpretable rules: E.g., “High density + no road access = high vulnerability”
- Identify vulnerable zones: Classify areas as Low, Medium, or High risk
- Integrates outputs from Modules 2–4 for final decision-making

REQUIREMENTS

1. **Hardware Requirements**
   - Minimum 12 GB RAM for image processing and model execution
   - CPU Intel Core i5 / AMD Ryzen 5 or higher.
   - NVIDIA GPU with CUDA support (≥ 6 GB VRAM recommended)
   - Minimum 50 GB free disk space (SSD preferred)
2. **Software Requirements**
   - Windows 10 / 11 (64-bit) or Ubuntu 20.04 or later
   - Python 3.x development environment
   - TensorFlow for AI model development
   - OpenCV for satellite image processing
   - NumPy for feature computation
   - GIS tools for spatial visualization

EXPECTED OUTCOME

- Automated detection of built-up areas and road networks from satellite imagery
- Identification of urban, semi-urban, and non-urban regions based on spatial features
- Analysis of settlement patterns to distinguish planned and unplanned areas
- Integration of environmental factors such as vegetation and water bodies
- Generation of urban vulnerability maps highlighting high, moderate, and low-risk zones

CONCLUSION

This project presents a transparent AI-based approach for assessing urban vulnerability using satellite imagery. By analyzing physical urban patterns and environmental indicators at a patch level, the system classifies regions into low, medium, and high risk using clear rule-based logic. The proposed solution supports urban analysis and prioritization while remaining interpretable, scalable, and suitable for decision-support applications.

THANK YOU